



**FACULTY OF SCIENCE – UNIVERSITY OF PERADENIYA**  
**DEPARTMENT OF STATISTICS AND COMPUTER SCIENCE**  
**END SEMESTER EXAMINATION (SEMESTER I) – 2020/2021**  
**ST1031 – STATISTICS APPLICATIONS I (1 CREDIT)**

Time allowed: **TWO Hours**

Answer **ALL** questions

**INSTRUCTIONS:** You should prepare a text document (MS – WORD or Writer) including all the necessary outputs, answers and relevant interpretation. Save the file with the name **XXXXXX.doc** or **XXXXXX.odt** where **XXXXXX** represents your registration number (Eg: If the registration number is “S/19/001” then the file name should be “**S19001.doc**” or “**S19001.odt**”)

Also save the R-studio project with the relevant R-scripts with the name “**XXXXXX.Rproj**” in your home directory. **XXXXXX** represents your registration number. Inside this project, create a separate R-script file for each question including all the R commands and save your R-script files with the file names below.

|               |        |        |        |        |
|---------------|--------|--------|--------|--------|
| Question      | 1      | 2      | 3      | 4      |
| R-script file | Que1.R | Que2.R | Que3.R | Que4.R |

You are advised to save all your work regularly to avoid losses of data due to power failures.

1. The file **Housing.csv** in your home directory contains information collected by the U.S Census Service concerning housing in the area of Boston in 2022. The dataset contains twelve variables as follows.

| Variable         | Description                                                   |
|------------------|---------------------------------------------------------------|
| Price            | Estimated selling price of the house (\$)                     |
| Area             | Area of the house in square feet (sqft)                       |
| Bedrooms         | Number of bedrooms in the house                               |
| Bathrooms        | Number of bathrooms in the house                              |
| Stories          | Number of stories in the house                                |
| Mainroad         | Whether the house is located in a main road or not            |
| Guestroom        | Availability of the guestroom (Yes/No)                        |
| Basement         | Availability of the basement (Yes/No)                         |
| Hotwaterheating  | Availability of the hot water (Yes/No)                        |
| Airconditioning  | Availability of the air conditioning (Yes/No)                 |
| Parking          | Number of cars that can be parked in the house                |
| Prefarea         | Houses with a preferred neighborhood (Yes/No).                |
| Furnishingstatus | Whether the house is furnished, semi furnished or unfurnished |

- i. (10 Points) Import the **Housing.csv** file to your R working directory. Remove the columns **Mainroad**, **Hotwaterheating**, **Parking** and **Prefarea** and save the dataframe as **HousePricing**.
- ii. (15 Points) Obtain the summary statistics for variables: **Price**, **Area**, **Bedrooms** and **Bathrooms**, and interpret the results.
- iii. (10 Points) Using a suitable R command change the column names **Airconditioning** and **Furnishingstatus** to **Air\_conditioned** and **Furnishing\_status**, respectively.

- iv. (10 Points) Print the details of all houses in which the price is less than \$3000000 and the size is less than 4000 sqft, and store the results in a new dataframe named **Medium**.
  - v. (10 Points) Create a contingency table for the two categorical variables **Air\_conditioned** and **Guestroom**, and interpret the results.
  - vi. (15 Points) Add a new column named **Price\_per\_sqft** to **HousePricing**, where  $Price\_per\_sqft = Price/Area$ . Sort the dataset in increasing order of the variable **Price\_per\_sqft**.
  - vii. (15 Points) Draw a horizontal boxplot for **Price** according to the **Furnishing\_status**. Use the colors: green, yellow and blue for furnished, semi-furnished and unfurnished, respectively. Give a suitable title and axis names, and add a notch to the boxes on the boxplot.
  - viii. (15 Points) Find the frequency of stories and create a pie chart for the same variable using heat colors. Give a suitable title and display the number of stories and the percentage frequency as the labels of the pie chart.
2. a) The following table shows the number of sales that happened in three branches of a mobile phone company for a four month time period.

|         | January | February | March | April |
|---------|---------|----------|-------|-------|
| Branch1 | 100     | 112      | 206   | 125   |
| Branch2 | 287     | 134      | 98    | 105   |
| Branch3 | 148     | 212      | 111   | 200   |

- i. (10 Points) Create a matrix using the above data and name it as **S**. Name the columns and rows accordingly.
  - ii. (10 Points) The number of sales made by Branch 2 in February was mistakenly added as 134 instead of 143. Correct this data point in the matrix.
  - iii. (10 Points) Create a  $3 \times 4$  matrix **J** in which all the elements equal to one.
  - iv. (20 Points) Using the two matrices **S** and **J**, find the total sales of each branch and the total sales of each month, separately.
- b) A set of algebraic formulas can be written using matrix notations as follows:

$$\begin{pmatrix} ax + by \\ cx + dy \end{pmatrix} = \mathbf{AZ},$$

where  $\mathbf{A} = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$  is a constant matrix and  $\mathbf{Z}^T = (x \ y)$  contains the variables.

- i. (10 Points) Find the matrices **A** and **Z** corresponding to the following set of algebraic formulas.

$$\begin{pmatrix} x + 2y + 3z \\ -2y + z \\ -2x - 2z \end{pmatrix}$$

- ii. (10 Points) Is the matrix  $A$  obtained in part (i) invertible? If so, Find the inverse of the matrix  $A$ .
  - iii. (10 Points) Copy the elements of matrix  $A$  in part (i) to another matrix named  $B$ . Add 4 to all the elements in the second row of  $B$  and subtract 1 from all elements in the third row of  $B$ .
  - iv. (10 Points) Find the matrix  $D = A^T B$  and its transpose ( $D^T$ ).
  - v. (10 Points) Show that  $B \times B \times B$  is a scalar multiple of the identity matrix, and find the scalar.
3. a) Create the below given sequence of vectors using the functions **seq** and **rep** when necessary.
- i. (10 Points)  $V1 = \text{odd numbers from 1 to 21}$
  - ii. (10 Points)  $V2 = \{2,5,8, 2,5,8, 2,5,8,2,5,8\}$
  - iii. (10 Points)  $V3 = \{1.25, 1.25, \dots, 1.25, 1.5, 1.5, \dots, 1.5, 1.75, 1.75, \dots, 1.75\}$ , where there are 8 occurrences of 1.25, 5 occurrences of 1.5 and 4 occurrences of 1.75.
  - iv. (10 Points)  $V4 = \{0.1^3 * 0.2^1, 0.1^6 * 0.2^4, 0.1^9 * 0.2^7, \dots, 0.1^{36} * 0.2^{34}\}$
  - v. (10 Points)  $V5 = \left\{2, \frac{4}{2}, \frac{8}{3}, \dots, \frac{2^{25}}{25}\right\}$ .
- b) (5 Points) Create two data frames named **Reg\_Details** and **Results** and fill them with the following observations.

**Reg\_Details:**

| Reg_no | Name    | Gender |
|--------|---------|--------|
| 10     | Emili   | F      |
| 09     | Afliya  | F      |
| 08     | Nuwan   | M      |
| 07     | Hashani | F      |
| 06     | Danuki  | F      |
| 05     | Rasika  | M      |
| 04     | Sahani  | F      |
| 03     | Banuka  | M      |
| 02     | Anuka   | M      |
| 01     | Amal    | M      |

**Results:**

| Reg_no | Score | Attempts | Qualify |
|--------|-------|----------|---------|
| 01     | 8     | 1        | yes     |
| 02     | 7     | 2        | yes     |
| 03     | 5     | 3        | yes     |
| 04     | 5     | 1        | yes     |
| 05     | 1     | 2        | no      |
| 06     | 10    | 2        | yes     |
| 07     | 6     | 1        | yes     |
| 08     | 8     | 3        | yes     |
| 09     | 4     | 2        | no      |
| 10     | 4     | 3        | no      |

- i. (10 Points) Merge the two data frames and save it as **Exam\_Details**.
- ii. (5 Points) Select all Qualified candidates from **Exam\_Deatils** and create a subset with a suitable name
- iii. (10 Points) Find the Percentage of female candidates who are qualified from the test.
- iv. (10 Points) Extract the names of male candidates who are qualified in their first attempt.

- v. (10 Points) Find the mean and standard deviation of scores of male and female candidates, separately. Interpret your results.
4. a) A local electricity distribution company charges from its customers according to the table given below.

| Electricity consumption<br>(kWh) | Unit charge<br>(Rs/kWh) | Surcharge amount<br>(Rs) |
|----------------------------------|-------------------------|--------------------------|
| 1 to 100                         | 0.50                    | 0.00                     |
| 101 to 200                       | 1.20                    | 35.00                    |
| Over 200                         | 1.50                    | 55.00                    |

Then the bill amount of a particular customer is calculated as,

$$\text{Bill\_amount} = \text{Charges\_for\_consumed\_units} + \text{total\_surcharge\_amount},$$

where *total\_surcharge\_amount* is equal to the sum of all the surcharge amounts up to the category to which the bill amount belongs.

- i. (40 Points) Write an R-function to find the bill amount of a customer with the number of units consumed, using the two input arguments of the previous electricity meter reading ( $n$  kWh) and the current meter reading ( $m$  kWh). The function should return the bill amount rounded to the nearest two decimal digits.
- ii. (10 Points) Using the R-function created in part (i), find the bill amount of a customer whose previous electricity meter reading is 165233 and the current meter reading is 165338.
- b) (50 Points) Write a recursion function to find the sum of the first  $n^{\text{th}}$  terms of the sequence given below.

3, 17, 87, 437, ...

Using the function you created, find the sum of the first 10 elements of the sequence.

**- End -**